Jake Weinstein

jakefweinstein@gmail.com | 312-508-9438 | github.com/JakeWeinstein

Education

Northwestern University, Dual Degree: BS in Applied Mathematics, BS in Computer Science — September 2023 – June 2027

- GPA: 3.95/4.0
- Relevant Coursework: Differential Equations, Numerical Linear Algebra, GPU/Cuda Programming, MPI Parallel Programming, Probability, Computer Systems, Databases, Computational Differential Equation Modeling, Complex Analysis, AI/Machine Learning.

University of Edinburgh, College of Science and Engineering — September 2025 – December 2025

- Computer Science study abroad exchange program.

Work Experience

Trading Intern, Milliman – Chicago, IL — June 2026 - August 2026

- Developing a machine-learning-based hedging engine to dynamically hedge fixed indexed annuity (FIA) liability books.
- Designed a custom deep hedging architecture that learns hedging strategies end-to-end from simulated market paths, moving beyond hand-tuned Greek-based hedges.
- Applied attention-based neural networks to capture the path-dependence and cross-asset dynamics of the hedging problem.

Sailing Coach, Columbia Sailing School – Chicago, IL — June 2024 - August 2024; June 2025 - August 2025

- Full-time sailing coach teaching advanced Club 420 sailing and training the Club 420 Race Team.
- Designed and ran on-the-water drills and on-land "chalk-talks" to improve sailors' boat handling, speed, and racing tactics.
- Analyzed video footage of sailors' performance to provide targeted critique and feedback.

Research Experience

Research Assistant, Northwestern University – Prof. Niall Mangan's group — 2025 – Present

- Built a generalizable finite-element solver framework in Python, with AI coding agents (Claude Code), to robustly simulate a broad class of strongly-coupled nonlinear electrochemical systems and help uncover their unknown underlying reaction mechanisms.
- Developed a PDE-constrained calibration pipeline to recover unknown kinetic parameters from experimental data, pairing adjoint gradients (Pyadjoint) with a PyTorch neural surrogate for fast parameter inference.
- Leveraged AI coding agents (Claude Code) to build and manage a large research codebase – 113k+ lines of Python across 273 source files, and a 1,000+-test regression suite spanning solver development, numerical verification, and documentation.
- Engineered custom multi-agent AI workflows for scientific computing: parallelized literature research, multi-model independent correctness verification, and an adversarial multi-model critique loop to plan, implement, and validate numerical methods.

Extracurriculars

Northwestern University Sailing Team — September 2023 - Present

- Serve as captain: lead practices, organize regattas, mentor new members, and foster a welcoming team culture.
- Devise and run a winter training program to keep sailors engaged and improve team performance in the off-season.
- Competed as Northwestern's "A" skipper in 20+ regattas, including College Sailing Nationals in 2024 and 2026.

Skills

- **Programming:** Python, Matlab, Java, C++, C, SQL.
- **ML & Scientific Computing:** PyTorch, Firedrake, deep learning, attention networks, finite element methods, PDE-constrained/adjoint optimization.
- **Tools:** Claude Code, Codex, LaTeX, Git.